Politecnico di Milano
Dipartimento di Elettronica, Informazione e Bioingegneria
prof. Fabrizio Ferrandi

Parallel Computing–I part–Thursday, June 30th, 2023

Polimi ID ________ ____________________________

Surname ________ ______________________ **Name** _________________________

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:15m.

Exercise 1 (4 points) _________________________

Exercise 2 (4 points) _________________________

Exercise 3 (4 points) _________________________

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Exercise n. 1

Answer the following questions about PRAM models and briefly explain (without an explanation, the answer will be considered invalid)

A. Describe which type or read and write you have in a PRAM machine (1)

B. Give a definition of the speedup when a PRAM model is considered. (1)

C. Is there a way to estimate the memory usage of an algorithm when a PRAM model is considered? (1)

D. Gustafson's model is a serial-fixed-time model. True/False? (1)

Exercise n. 2

Answer the following questions about different forms of parallel executions and briefly explain (without an explanation, the answer will be considered invalid).

A. Describe the difference between the Message passing and Shared address space programming model. (1)

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B. Can a program with many arithmetic operations and a small number of memory accesses be a problem for a multi-threaded processor? True/False. Why? (1)

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C. Explain the difference between bandwidth and latency when a memory subsystem in a multithreaded processor is considered. (1)

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D. Is SIMD relevant for the GPUs? (1)

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Exercise n. 3

Answer the following questions about programming models plus CUDA and briefly explain (without an explanation, the answer will be considered invalid).

A. Describe the CUDA threading model? (1)

B. What is meant by "warp"? Is it part of the CUDA language? (1)

C. Describe the CUDA memory hierarchy. (2)

Exercise n. 4

Answer the following questions about memory and heterogeneous systems and briefly explain (without an explanation, the answer will be considered invalid).

A. Please show how the DRAM bursting is working. (2)

B. Please explain the characteristics of the sequential consistency model. (1)

C. How specialized architectures are relevant for heterogenous computing? (1)